MUDITHA PRIYASAD

Mechanical Design | Industrial Automation | R&D Engineer
BSc (Hons) Mechanical Engineering | IESL Associate Member AM-32795
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Detailed Professional CV | Engineering Experience, Projects and Credentials

Scan for Portfolio

PROFESSIONAL PROFILE

Mechanical Design, Industrial Automation and R&D Engineer with hands-on engineering exposure since 2019 and more than two years of industry delivery across 15+ projects. Experience covers machine and fixture design, rapid prototyping, PLC-based control, pneumatics, commissioning, troubleshooting, maintenance improvement and practical engineering systems. Known for low-cost solutions that improve throughput, repeatability, operator effort and production support within real workshop and factory constraints.

TECHNICAL CAPABILITY

- **Mechanical design:** mechanisms, parts and assemblies, jigs and fixtures, 3D CAD, DfMA, fabrication support and design validation.
- **Automation and controls:** PLC programming/support, HMI troubleshooting, sensors, actuators, pneumatics, sequencing, commissioning, wiring awareness and fault finding.
- **Manufacturing and maintenance:** root-cause analysis, TPM, 5S, Kaizen, SOPs, KPI tracking, low-cost automation, production support and reliability thinking.
- **Rapid prototyping:** FDM printing, PETG/ABS/ASA material selection, design for printing, printed tooling, CNC laser work and iterative fit checks.
- **Engineering software:** SOLIDWORKS, Siemens STEP 7 Micro/WIN, Delta ISPSoft, Mitsubishi GX Works, Xinje PLC Software, MS Excel, OrcaSlicer, Cura, LightBurn, Inkscape, Arduino IDE, Python, C++ and Raspberry Pi.
- **Additional familiarity:** AutoCAD, MATLAB and Power BI.
- **Professional strengths:** collaboration, continuous improvement, design thinking, documentation, engineering communication and practical problem solving.

MEASURABLE ENGINEERING HIGHLIGHTS

- Raised fan-base filling throughput from 1 to 6 bases per minute through a semi-automatic guided filling and assembly machine.
- Raised B22 eyelet-punching output from 3-4 to 16 pieces per minute while reducing operator effort.
- Developed a hybrid 3D-printed and laser-cut stainless-steel motor fixture that saved about 5 seconds per motor across 50-80 motors per day.
- Designed a prototype 1W LED bulb assembly machine with capacity up to 20 pieces per minute.
- Delivered 20+ MTone custom jobs and built/tested 3 custom 3D printers and 4 laser machines.
- Built a PCB drilling system that handles 18 PCBs in one set-up through stacked positioning.

PROFESSIONAL EXPERIENCE

MARQO Industries | Mechanical Design & Automation Engineer

Dec 2024 - Present

- Appointed as the company's first engineer and established the engineering function from the ground up, including workshop organization, tooling control, engineering inventory, practical documentation, and supporting software workflows.
- Converted production needs into maintainable machines, jigs, fixtures, and pneumatic assemblies through concept development, CAD detailing, component selection, fabrication follow-up, commissioning, operator feedback, and iterative improvement.

- Set up, commissioned, and troubleshot spot-welding, packing, pad-printing, and LED-cup installation equipment, resolving mechanical, pneumatic, alignment, and process issues during production use.
- Developed production solutions including a fan-base machine, eyelet puncher, LED-bulb prototype, and fan-motor assembly fixture, with attention to repeatability, safe operation, ease of adjustment, and local manufacturability.
- Provided day-to-day production engineering support through fault isolation, mechanism adjustment, pneumatic repair, fabrication coordination, operator guidance, and follow-up improvements that reduced recurring interruptions.

MTone Designers | Founder

- Delivered more than 20 customized jobs across FDM 3D printing, laser cutting and engraving, personalized products, and engineering applications, managing requirements, design preparation, manufacturing, assembly, inspection, and customer handover.
- Built, wired, calibrated, tested, and improved three custom 3D printers and four laser machines, integrating mechanical frames, motion systems, electronics, control settings, and safety-focused operating procedures.
- Designed practical production aids including a PCB drilling fixture capable of holding 18 boards per setup, a pneumatic wire-stripping device, and multiple 3D-printed or laser-cut jigs, enclosures, guides, and replacement parts.
- Developed the MTone ERP desktop application to manage customer jobs, quotations, inventory, material and machine costing, production records, and operational follow-up in one maintainable workflow.

MAS Active Shadowline | Engineering Intern

Apr 2024 - Nov 2024

- Supported factory engineering improvement through Total Productive Maintenance, 5S, Kaizen activities, KPI tracking, maintenance checklists, process standardization, and coordination with production and technical teams.
- Served as Interim Sustainability Executive alongside automation duties, supporting practical resource-efficiency and workplace-improvement activities while maintaining engineering follow-up responsibilities.
- Prepared a coded machine-gauge reference standard covering flat-seam, overlock, single-needle, zigzag, flat-lock, special, and magnetic gauge groups to improve identification and consistency during machine setup and maintenance.
- Developed a phased proposal and layout for a six-section service center covering inspection, part replacement, disassembly and cleaning, finishing, reassembly and final checks, and documentation; planned machine flow, component lists, quotations, special tools, electrical and compressed-air distribution, and contractor coordination.
- Contributed to machine and process standards, repair-workshop layout improvements, and the Align Tank automation initiative, focusing on safer handling, clearer workflows, maintainability, and reduced equipment downtime.

Effective Engineering & Automation | Automation Engineering Intern

Feb 2023 - Aug 2023

- Supported machine design, fabrication, commissioning, troubleshooting, preventive and corrective maintenance, and field servicing across tensile-testing equipment, concrete mixers, lubricant-filling systems, glove-production machinery, screen-printing equipment, and RFID hardware.
- Managed an end-to-end custom FDM 3D-printer initiative covering mechanical architecture, aluminium-extrusion fabrication, component procurement, electrical wiring, BTT SKR 1.4 and Marlin configuration, calibration, fault correction, and successful machine and print validation.

- Built practical controls experience by diagnosing and monitoring Haiwell and Xinje PLC systems, checking relay/NPN/PNP inputs and expansion modules, testing focused ladder routines, completing industrial three-phase control wiring, and improving a Haiwell SCADA interface for real-time process monitoring.
- Designed or developed concepts and mechanisms for variable-volume liquid filling, pneumatic screen printing, coconut-oil production feeding, a compact self-tightening gripper, and an oil-recovery drip tray; prepared drawings, selected components, supported assembly, and refined designs through testing.
- Participated in workshop and site work including lathe and fabrication tasks, off-grid solar-panel installation, pelletizer testing, sensor installation and calibration, equipment breakdown response, technical documentation, and operator/user guidance.

MAS Kreeda Vaanavil | Autonomation Engineering Intern *Jan 2019 - Apr 2019*

- Identified a manual bonding-tape application bottleneck and helped translate it into an automated machine sequence covering tape feeding, variable-length measurement, cutting, transfer, heating, bonding, and fabric return.
- Contributed to the mechanical, pneumatic, electrical, and control integration of the Ottuthal bonding-tape machine using a PLC, stepper drive, PID temperature control, SSR, thermocouple, cartridge heaters, suction, solenoid valves, cylinders, and linear-motion components.
- Supported iterative testing and problem solving for programmable tape length, temperature stability, cylinder-completion sensing, tape positioning, fabric grip, and two-hand start protection, improving repeatability, maintainability, and operator safety.
- Applied CAD, wiring, pneumatics, Arduino, and PLC-ladder fundamentals while researching next-stage fabric separation and handling concepts using airflow/Bernoulli effects and garment stretch properties, and contributing to a remote boiler-monitoring concept.

COMPLETE PROJECT RECORD

This record combines documented work from employment, internships, university projects and MTone Designers. Employer-sensitive drawings, PLC logic, wiring, customer data and exact production details are intentionally excluded.

1W LED Bulb Assembly Machine | *Industrial R&D / Automation*

Designed and built the prototype mechanical layout, actuation and assembly sequence using sensors, pneumatics and control logic. Demonstrated a semi-automatic end-to-end concept with capacity up to 20 pieces per minute and reduced repetitive handling.

Fan Motor Assembly Fixture | *Jig & Fixture Design*

Modelled holding and locating features in SOLIDWORKS and iterated a hybrid 3D-printed and laser-cut stainless-steel design. Saved about 5 seconds per motor across 50-80 motors per day while improving handling and repeatability.

Fan Base Filling Machine | *Industrial Automation*

Developed and commissioned a guided pneumatic filling and assembly cycle to replace slow, inconsistent manual work. Raised throughput from 1 to 6 bases per minute with operator loading and refilling.

B22 Eyelet Puncher Machine | *Machine Build / Automation*

Worked on the punching mechanism, feeding, pneumatic actuation and operator-safe sequence. Raised output from 3-4 to 16 pieces per minute and improved repeatability and handling.

CNC Laser Engraver | *Machine Build / Digital Manufacturing*

Designed the frame and motion system, completed wiring and configured stepper control, GRBL and LightBurn. Built a working low-cost engraver for customised products and process trials.

Custom FDM 3D Printers | *Machine Build / Rapid Prototyping*
Built frames and motion systems, wired electronics, configured firmware and tuned slicing and print quality. Built and tested 3 printers for jigs, fixtures, mechanisms, prototypes and customer parts.

Bio-Inspired Hopping Robot | *Final-Year Research / Robotics*
Designed and fabricated a two-actuator hopping mechanism inspired by grasshopper jumping. Worked on actuator selection, control and prototype tuning and demonstrated a working hopping concept.

Combat Robots | *Robotics / Team Leadership*
Led the university team and worked on drivetrain, armour, weapon mechanisms, fabrication, electronics and testing. Entered working robots in UWU Robot Battles 2018/2019 and SLT Robot Battles 2019.

Pneumatic Wire-Stripping Machine | *Automation / Production Support*
Developed a stripping mechanism and pneumatic cycle to replace slow, inconsistent manual stripping. Improved consistency and reduced handling time for customised work.

CNC PCB Drilling Machine | *Machine Build / CNC*
Built compact CNC mechanics, stepper motion and drilling control with stacked PCB positioning. Enabled 18 PCB pieces per set-up with better alignment, repeatability and finish.

Bonding Tape Attaching Machine | *Internship Automation Project*
Worked on PLC sequence concepts, PID temperature control, pneumatics, sensors and stepper-based feeding for an automated tape feeding, measuring, cutting and heat-bonding cycle.

8x8x8 LED Cube | *Electronics / Embedded Project*
Designed the layout, hand-soldered the LED matrix and programmed microcontroller logic and animations. Completed a working animated cube demonstrating electronics, programming and build discipline.

Line-Follower Robot with Liftable Gripper | *Robotics / Embedded Control*
Built a sensor-guided line-following platform and liftable gripper to demonstrate autonomous movement and a simple pick-and-place action.

Align Tank Standardisation | *Manufacturing Improvement*
Standardised sewing-machine settings for a given style across a production line and supported process documentation, improving set-up consistency and reducing changeover errors.

TPM Implementation Project | *Maintenance / Continuous Improvement*
Supported Fuguai tagging, autonomous-maintenance training, operational checks and process standardisation to strengthen equipment ownership and maintenance discipline.

MTone ERP Software | *Engineering Operations Software*
Developed a custom desktop ERP workflow for customer and job records, quotation, inventory and costing, creating a practical operating system for consistent business control.

INDUSTRIAL MACHINE & PROCESS EXPOSURE

- Production and assembly: spot welding, packing, pad printing, LED-cup installation, fan-base filling, B22 eyelet punching and LED-bulb assembly.
- Service and troubleshooting: tensile-testing, concrete-mixer, oil-filling, glove-making and screen-printing machines.
- Automation technologies: PLC/HMI support, sensors, actuators, pneumatics, stepper motion, control logic, PID concepts and fault isolation.
- Digital manufacturing: custom FDM printers, CNC laser systems, PCB drilling, slicing, print tuning, LightBurn/GRBL and rapid prototyping.

- Manufacturing systems: TPM, 5S, Kaizen, SOPs, KPI tracking, operational checklists, workshop layout and machine/process standardisation.

EDUCATION & RESEARCH

University of Jaffna | BSc (Hons) Mechanical Engineering

Oct 2017 - Feb 2024

- Final-year research: designed and fabricated a two-actuator hopping robot with grasshopper-inspired jumping mechanisms.
- Led the combat-robot team, built an 8x8x8 LED cube and held leadership roles in practical engineering projects and exhibitions.

PROFESSIONAL CREDENTIALS & TRAINING

- IESL Associate Member - Institution of Engineers Sri Lanka | AM-32795.
- Certified SOLIDWORKS Associate (CSWA) - Mechanical Design | Dassault Systemes SOLIDWORKS.
- Certified SOLIDWORKS Associate (CSWA-AM) - Additive Manufacturing | Dassault Systemes SOLIDWORKS.
- PLC Training - Robotics & Control Systems Laboratory, University of Moratuwa | 2025.
- PLC Programming Course - Epic Engineering | 2024.
- Maintenance of Industrial Plant & Machinery - NERDC | 2026.
- Industrial Motor Operation & Control - NERDC | 2025.
- Advanced Injection Moulding Technology - Industrial Development Board of Ceylon | 2026.

ACHIEVEMENTS & PROFESSIONAL ACTIVITIES

- Bronze Award / 3rd place for the University of Jaffna stall with the best engineering project display at Techno Sri Lanka 2023.
- Participant: Air Tattoo Exhibition 2024, Techno Sri Lanka 2023 and SLT Robot Battles 2019.
- Organiser: University of Jaffna Battlegrounds 2020.
- Project and innovation exposure through university exhibitions, robot competitions and engineering demonstrations.

LANGUAGES

- English - Professional proficiency.
- Sinhala - Native.

CONFIDENTIALITY & EVIDENCE NOTE

Public-safe photographs, videos, certificates and evidence are available through the portfolio. Employer drawings, CAD files, PLC/HMI source, wiring diagrams, customer information, production data and factory details remain confidential and are discussed only with permission.